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Client

Gujarat Fluorochemicals Ltd
12/A GIDC Industrial Estate, Dahej,
Bharuch, Gujarat 392130

Earthing System Health Assessment

FI Plant · Final Report

Revision	Date of Submission	Issue	Prepared By	Checked By	Approved By
R0	dd-mm-yyyy	First Issue	HSP	AP	JCS

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The index lists only the tests performed for this unit. Sections are generated from the site records.

1. Executive Summary

A. Overall summary of tests

Sr.	Test	Unit	Total	Within limit	Above limit	Not assessed	Pass %	Overall status
1	Earth Electrode Resistance	pits	10	8	1	1	88.9%	>Attention Required
2	Earth Fault Loop Impedance	panels	4	3	1	0	75.0%	>Attention Required
3	Prospective Fault Current	panels	4	3	0	1	100.0%	Healthy
4	Riser Integrity	points	6	5	1	0	83.3%	>Attention Required

Pass % = points within limit ÷ assessed points (Warning-band points are within limit but carry priority P2). Overall status: 0 unhealthy = Healthy; pass ≥ 90% = Healthy – Minor Actions; otherwise >Attention Required (ET-D03-REG-002, OVERALL).

B. Earth electrode priority summary

Priority	Meaning	Qty	Pits
P1	Exceeds limit, not measurable, or safety defect – rectify within 30 days and re-test	3	EP-03, EP-07, EP-08
P2	Warning band or maintenance defect – rectify in maintenance cycle	2	EP-05, EP-09
P3	Healthy – routine annual testing	5	EP-01, EP-02, EP-04, EP-06, EP-10

C. Key findings

• Soil resistivity at FI Plant averages 25.7 Ω·m, classified moderately corrosive per IS 3043:2018 Table 4; corrosion allowance applies to steel earthing conductors.
• 3 of 10 earth pits require priority action: EP-07 exceeds its calculated limit with the strip disconnected, EP-03 is heavily corroded although within limit, and EP-08 could not be measured because the chamber is buried.
• Feeder 7F2 at MCC-2 exceeds the 1.8 Ω loop impedance criterion on the B phase; the other three panels are healthy.
• Riser R-202 (R1) reads 1.42 Ω towards the equipment, above the 1 Ω criterion, with a loose connection noted; EP-09 is within limit but above 90% of it and is flagged P2 for monitoring.
• APDB-2 prospective fault current cannot be assessed because the MCB breaking capacity was not recorded.

Findings text is written by the report AI from the computed results above and checked by the sanity check before review; it cannot change any value or status.

D. Overall condition

The earthing system of the FI Plant is in fair condition. Three earth pits, one feeder and one riser point need corrective action before the next annual test; all other measured points meet the acceptance criteria stated in Section 3.

2. Introduction & Scope of Work

Item	Detail
Client	Gujarat Fluorochemicals Ltd
Site address	12/A GIDC Industrial Estate, Dahej, Bharuch, Gujarat 392130
Unit	FI Plant
Survey dates	14-09-2026, 15-09-2026 (2 site days)
ETPL engineers	H. S. Patil, A. S. Mahto
Client representative	Mr. B. Thakkar

Scope – planned vs completed

Test	Unit	Planned (Zoho task)	Completed	Remark
Soil resistivity	locations	2	2	
Earth electrode resistance	pits	10	9	1 not measurable (EP-08)
Loop impedance & PFC	panels	4	4	
Riser integrity	points	6	6	

3. Basis of Assessment

Every limit, formula and classification used in this report is listed below with its source. Values are taken from the project's acceptance profile (ET-D03-REG-002, profile ETPL-DEFAULT). Criteria marked ETPL acceptance criterion are ETPL's adopted values where the cited standard sets no number.

Criterion	Rule / value used	Basis	Source & clause
Method and calculation	Wenner 4-probe, probe depth 10-15 cm, rho = 2*pi*a*R	Standard	IS 3043:2018, Cl. 40.3, 40.4.1
Classification	Corrosivity: <25 severely; 25-50 moderately; 50-100 mildly; >100 very mildly corrosive. No pass/fail.	Standard	IS 3043:2018, Cl. 13.6.1, Table 4
Standard limit - rod/pipe	R = (100*rho / (2*pi*l)) * ln(2*l/d), l and d in cm	Standard	IS 3043:2018, Cl. 14.2.2
Standard limit - strip/conductor	R = (100*rho / (2*pi*l)) * ln(4*l/d), d = strip width or 2x conductor diameter, cm	Standard	IS 3043:2018, Cl. 14.2.3
Measurement method	Fall of potential, potential probe at 61.8%; clamp-on, slope or 90/180 as alternatives	Standard	IS 3043:2018, Cl. 41, 41.1
Status bands	Healthy <= 0.9 x limit; Warning > 0.9 x limit and <= limit; >Permissible Limit > limit	ETPL acceptance criterion	ETPL practice, Cl. ET-D03 ESHA Standard 2.2
Priority	P1: > limit, not measurable, or C1/C2/C4/C5. P2: warning band or C3/C6/C7/C8. P3: healthy, no code.	ETPL acceptance criterion	ETPL practice, Cl. ET-D03 ESHA Standard 2.2, Part 3
Earth fault loop impedance Zs	Zs <= limit for all three phases (value: 1.8 ohm)	ETPL acceptance criterion	ETPL practice; actual device limit per IS/IEC 60364-4-41, Cl. IS 3043:2018 Cl. 42 (method)
Prospective fault current	PFC < protective device breaking capacity (kA); not assessable if kA not recorded	Standard	IS/IEC 60364-4-43, Cl. 434.5.1
Riser bonding resistance	Both towards-equipment and towards-grid <= limit (value: 1.0 ohm)	ETPL acceptance criterion	ETPL practice (IEEE 81-2012 Cl. 10 to be confirmed), Cl. -
Overall test status	0 unhealthy: Healthy; pass >= 90%: Healthy - Minor Actions; else >Attention Required	ETPL acceptance criterion	ETPL practice, Cl. ET-D03 ESHA Standard 2.10
Re-testing interval	Annual; never less than once in two years; records kept	Standard	IS 3043:2018, Cl. 38.4.1, 38.4.2

Soil resistivity used for electrode limits: ρ = 25.68 Ω·m (measured average of both locations, Section 6). No assumed values were used.

4. Instruments Used

Instrument	Model	Serial No.	Calibration cert.	Valid until
Earth analyser	Metrel MI 3290 GF	18480898	[cert no.]	[date]
Installation tester	Metrel MI 3152 EurotestXC	17500653	[cert no.]	[date]
Loop / PSC tester	Kyoritsu 4118A	E0182298	[cert no.]	[date]

5. Testing Methodology

Soil resistivity

Wenner four-probe method: four probes at equal spacing a in a straight line, driven 10–15 cm; current injected through the outer probes and voltage measured across the inner probes; ρ = 2πaR (IS 3043:2018 Cl. 40). Measured in two perpendicular directions at spacings of 0.5–3.0 m.

Earth electrode resistance

Fall-of-potential method with the electrode isolated from the grid, potential probe at 61.8% of the current-probe distance, repeated at ±10% to confirm a stable reading (IS 3043:2018 Cl. 41). Each result is compared with the electrode's calculated resistance from Cl. 14.2 using the measured soil resistivity and the recorded electrode dimensions.

Loop impedance & PFC

Live phase-to-earth loop impedance (R-E, Y-E, B-E) measured at each panel; prospective fault current measured phase-to-neutral and compared with the protective device's breaking capacity (IS/IEC 60364-4-43 Reg. 434.5.1).

Riser integrity

Low-resistance measurement from each riser point towards the equipment and towards the earth grid.

6. Soil Resistivity – Results

Location: Utility Yard

Spacing a (m)	ρ Direction 1 (Ω·m)	ρ Direction 2 (Ω·m)	Mean ρ (Ω·m)
0.5	30.79	32.67	31.73
1.0	28.90	30.79	29.85
1.5	27.33	29.22	28.27
2.0	25.13	27.65	26.39
2.5	23.56	25.13	24.35
3.0	22.62	24.50	23.56
Location average			27.36

Location: Reactor Bay

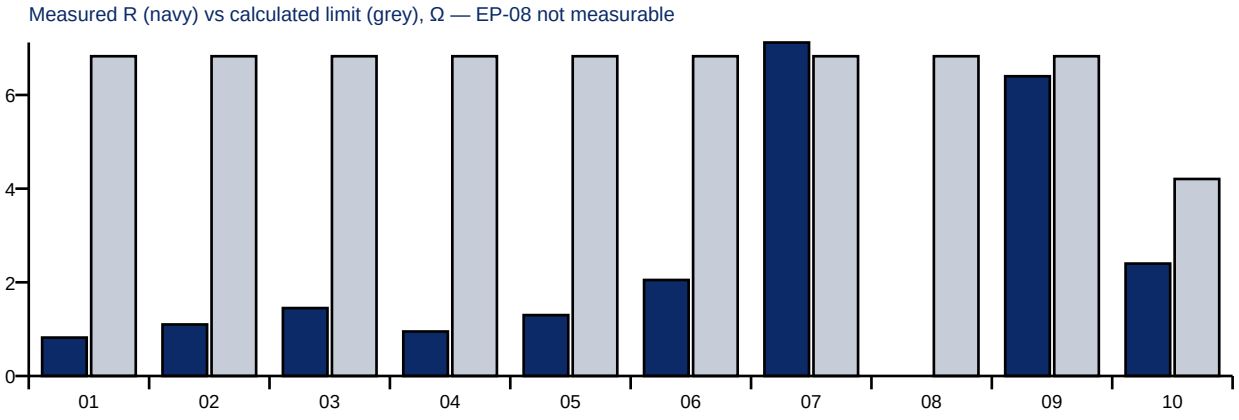
Spacing a (m)	ρ Direction 1 (Ω·m)	ρ Direction 2 (Ω·m)	Mean ρ (Ω·m)
0.5	25.45	27.02	26.23
1.0	24.50	25.76	25.13
1.5	24.50	25.45	24.98
2.0	22.62	23.88	23.25
2.5	21.99	23.56	22.78
3.0	20.73	22.62	21.68
Location average			24.01

Site average ρ (Ω·m)	Corrosivity class (IS 3043:2018 Table 4)	Implication
25.68	Moderately corrosive	Steel conductors: 15% corrosion allowance in cross-section (IS 3043:2018 Table 5); steel rods ≥ 16 mm dia, flats ≥ 6 mm thick (Cl. 13.6.1)

7. Earth Electrode Resistance – Results

Sr	Pit tag	Location	Type	l (cm)	d (cm)	Limit (Ω)	Meas. R (Ω)	Status	Codes	Priority
1	EP-01	Transformer TR-1	Rod	300	4.0	6.83	0.82	Healthy	—	P3
2	EP-02	Transformer TR-1	Rod	300	4.0	6.83	1.10	Healthy	—	P3
3	EP-03	MCC Room	Rod	300	4.0	6.83	1.45	Healthy	C1	P1
4	EP-04	MCC Room	Rod	300	4.0	6.83	0.95	Healthy	—	P3
5	EP-05	Reactor Bay R-201	Rod	300	4.0	6.83	1.30	Healthy	C8	P2
6	EP-06	Reactor Bay R-202	Rod	300	4.0	6.83	2.05	Healthy	—	P3
7	EP-07	Cooling Tower	Rod	300	4.0	6.83	7.12	>Permissible Limit	C2	P1
8	EP-08	Pump House	Rod	300	4.0	6.83	—	Not Measurable	C4	P1
9	EP-09	Pipe Rack PR-3	Rod	300	4.0	6.83	6.40	Warning	—	P2
10	EP-10	DG Area	Strip	600	5.0	4.21	2.40	Healthy	—	P3

Limits: rod R = (100ρ/2πl)·ln(2l/d) (IS 3043:2018 Cl. 14.2.2); strip R = (100ρ/2πl)·ln(4l/d), d = strip width (Cl. 14.2.3); ρ = 25.68 Ω·m. Healthy ≤ 90% of limit; Warning 90–100%.



8. Earth Fault Loop Impedance & PFC – Results

Panel	Feeder	Device	Rating (A)	Br. cap. (kA)	Zs R/Y/B (Ω)	Zs status	PFC R/Y/B (kA)	PFC status
PMCC-1	Incomer	ACB	1600	50	0.62 / 0.65 / 0.61	Healthy	6.90 / 6.70 / 7.10	Healthy
MCC-2	7F2 Feeder	MCCB	250	36	1.21 / 1.35 / 1.92	>Permissible Limit	2.10 / 2.00 / 1.40	Healthy
MLDB-1	Incomer	MCCB	125	25	1.05 / 1.10 / 1.02	Healthy	2.60 / 2.50 / 2.70	Healthy
APDB-2	Incomer	MCB	63	—	1.40 / 1.38 / 1.44	Healthy	1.90 / 1.90 / 1.80	Not Assessable

A panel is healthy only if all three phases meet the criterion. Zs criterion 1.8 Ω is an ETPL acceptance criterion (Section 3).

9. Riser Integrity – Results

Sr	Main location	Point	Towards equipment (Ω)	Towards grid (Ω)	Codes	Status
1	Reactor Bay	R-201 (R1)	0.21	0.12	—	Healthy
2	Reactor Bay	R-201 (R2)	0.34	0.20	—	Healthy
3	Reactor Bay	R-202 (R1)	1.42	0.31	C6	>Permissible Limit
4	Pump House	P-101 A	0.18	0.09	—	Healthy
5	Pump House	P-101 B	0.27	0.19	—	Healthy
6	Cooling Tower	CT Fan-1	0.44	0.38	—	Healthy

10. Corrective Action Plan

Sr	Test / point	Observation	Corrective action	Priority / status
1	Electrode EP-03	R = 1.45 Ω within limit 6.83 Ω. Rod heavily corroded at clamp.	Rectify within 30 days: repair/replace electrode and strip, restore test link and bonding (IS 3043:2018 Cl. 14.2, 25.6.3.4); re-test after rectification.	P1
2	Electrode EP-05	R = 1.30 Ω within limit 6.83 Ω. No pit tag.	Fix a permanent tag/label at the pit; no electrical action required.	P2
3	Electrode EP-07	R = 7.12 Ω exceeds limit 6.83 Ω. Strip disconnected from rod.	Rectify within 30 days: repair/replace electrode and strip, restore test link and bonding (IS 3043:2018 Cl. 14.2, 25.6.3.4); re-test after rectification.	P1
4	Electrode EP-08	Not measurable. Chamber buried under paving.	Expose and restore pit chamber with removable cover; measure and report within 30 days.	P1
5	Electrode EP-09	R = 6.40 Ω is above 90% of limit 6.83 Ω.	Rectify in the next maintenance cycle; monitor at next annual test.	P2
6	Loop MCC-2 7F2 Feeder	Zs above 1.8 Ω on at least one phase.	Check PE conductor terminations and size at the feeder; re-test. Confirm device disconnection time per IS/IEC 60364-4-41.	>Permissible Limit
7	PFC APDB-2	Breaking capacity not recorded.	Record device nameplate kA and re-assess against measured PFC (IS/IEC 60364-4-43 Reg. 434.5.1).	Not Assessable
8	Riser R-202 (R1)	Towards equipment 1.42 Ω above 1 Ω; loose connection noted.	Clean and re-tighten joint, re-test both directions.	>Permissible Limit

Generated from the same per-point records as Sections 1 and 7–9; counts cannot differ between sections.

11. Maintenance & Re-testing

Measure earth resistance annually and in any case not less than once in two years; keep records of initial and subsequent tests; carry out periodic visual inspection of all earth electrode connections for rigidity and deterioration (IS 3043:2018 Cl. 38.4.1–38.4.3).

12. Annexures

Annexure	Content
A	Signed field measurement sheets (scans), MS-01/02/04/05
B	Photographs of all P1 points, by pit tag
C	Instrument calibration certificates
D	Daily progress reports issued during the survey